

Nirvaan Shah

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EDUCATION

University of Toronto

2024 – Expected 2029

B.A.Sc. in Civil Engineering with PEY; Minors in Engineering Business and Environmental Engineering

Relevant Coursework: Structural Analysis, Reinforced Concrete, Steel and Timber Design, Water and Wastewater Treatment Processes, Sustainable Energy Systems, Urban Engineering Ecology, Construction Management, Python Computer Programming, Engineering Economics, Fundamentals of Accounting and Finance, Marketing & Competitive Strategy, Survey Camp

WORK EXPERIENCE

Engineering Project Analyst Intern *Evardi (Y Combinator F26)* | Remote, Part-Time Alongside Term **August 2026 – Present**

- Building network topology models for 3 utility grids (substations, lines, transformers, and renewables) to simulate outages and equipment failures and test system reliability under stress.
- Developing optimization and machine-learning models and agents in Python to analyze grid operating data, automate anomaly detection, and evaluate control strategies across 50,000+ system measurements and scenarios.

Sustainability and Supply Chain Intern *Testo* | Lenzkirch, Germany

June – August 2026

- Found a way to recover 15% more platinum across 12 production areas. Costed five compliance options ranging from €12,000 to €30,000 and recommended one that met the requirements at 30% below the highest-cost option, while flagging a serious problem with the disposal of platinum waste.
- Led four interns in auditing 30 hazardous-material and gas work areas for environmental compliance; assessed 220 regulated substances and 70 machines and identified 45 gaps before a regulatory inspection.

Engineering Consulting Intern *Samir Consulting* | Nairobi, Kenya

June – August 2025

- Supported a 6-person engineering team on the design of a golf course resort development across a 110-hectare site.
- Produced 4 renderings and 3 designs using AutoCAD, Fusion 360, and Revit.
- Surveyed 40 hectares with a 4-person crew to record actual ground conditions and verify the proposed layout against the terrain.

DESIGN TEAMS

Aerostructural Design Team Member *UTSM* | Toronto, ON

September 2024 – Present

- Designed 12 structural components for a fuel-efficient vehicle body in SOLIDWORKS for the Shell Eco-marathon; the team placed second and passed technical inspection.
- Fabricated 20 vehicle components using laser cutting, 3D printing, and power tools as part of a 6-person team.

PROJECTS

Design & Cost Analyst *Clarke Prize Environmental Design Challenge* | Toronto, ON

- Placed 3rd out of 70 teams in U of T Engineering's Clarke Prize, a three-day sustainable design competition, advancing through four presentation rounds before roughly 500 attendees to a 6-team final judged by an industry panel.
- Designed BoxDrop, a lobby cardboard-recovery station with a barcode scanner, a sealed intake, and a child-safe cutter. 74% of apartment residents say they want to recycle more; what stops them is inconvenience and a lack of incentives.
- Developed a financial plan covering \$100,000–\$150,000 in app development costs, \$1,500 per unit plus \$350 per unit annually in maintenance, and \$30,000 per year for servers; benchmarked the concept against Ontario's \$15 million Deposit Return Program.

Project Manager *ESP Engineering Team* | Toronto, ON

- Led a 6-person team to design a solar microgrid for the University of Toronto's sustainability office, generating 1.46 GWh annually and saving 1,171 tonnes of CO₂, achieving net-zero carbon by 2038.
- Built the energy model, accounting for panel efficiency, tilt, azimuth, and seven system losses, yielding an 81.23% performance ratio.

Modeling Lead *Hydrological Modeling of Bear Creek* | Toronto, ON

- Delineated the Bear Creek watershed from an OpenTopography DEM in ArcGIS and built a HEC-HMS model using SCS Curve Number losses, a Snyder unit hydrograph, and gridded temperature-index snowmelt.
- Calibrated the model against two Water Survey of Canada gauges, adjusting the Curve Number from 60 to 70 and initial baseflow from 3 to 0.5 m³/s to correct an underestimated flood peak.

Cost Estimator *BIM Cost Estimation of a Residential Build | Toronto, ON*

- Priced 208 RSMean line items against a Revit BIM model to produce a total construction estimate of \$144,630, selecting assembly- or unit-cost methods by component family.
- Developed a 2018–2025 cost-escalation method using Statistics Canada indices and proposed value-engineering alternatives projected to save \$6,739.

Analysis Lead *Canada Post: Organizational Behavior Analysis | Toronto, ON*

- Analyzed Canada Post's \$748 million net loss in FY2023 and identified drivers of declining employee motivation using expectancy and operant-learning theories.
- Proposed voluntary capped overtime and a five-stage Protected Recovery System to address the burnout and role overload identified through the Job Demands-Resources model.

Author *Nairobi Urban Metabolism Study, Individual Project | Toronto, ON*

- Quantified annual resource flows for Nairobi's 5.8 million residents, including 22.7 GJ of energy and 14.6 tonnes of clean-water inflow per person, compared with 29 tonnes of wastewater outflow per person.
- Proposed converting the Dandora dumpsite into a waste-to-energy facility with a capacity of 2,000 t/day, yielding 45–60 MW, and implementing decentralized wastewater treatment based on Kampala's DEFAST model.

GIS Analyst *Fontas River Watershed Delineation | Toronto, ON*

- Processed a 30 m DEM in ArcGIS Pro through fill, flow direction, flow accumulation, and stream reclassification to delineate a boreal watershed in northeastern British Columbia.
- Extracted a 10,039 km² watershed polygon and checked it against BC government and iMapBC data. It came out 8% high, traced to the flat muskeg along the divide, where a 30 m DEM can't tell which way water drains.

Research Lead *Wastewater Reusability | Toronto, ON*

- Compared municipal wastewater systems in Curitiba, Shanghai, Markham, and Nairobi, contrasting linear collect-treat-discharge models with circular resource-recovery systems.
- Recommended anaerobic digestion, biogas recovery, and non-potable water reuse to reduce Markham's projected emissions of 233,000 tonnes of CO₂e.

LEADERSHIP & VOLUNTEERING

Founding Member and Advisor *Still I Rise | Nairobi, Kenya***March 2023 – Present**

- Co-founded the first tuition-free IB school serving the Mathare community in Nairobi over an 18-month period.
- Tutored 60 students in four subjects, mathematics, physics, chemistry, and Python, and helped secure \$50,000 in funding and sponsorships.
- Now advise on sponsor outreach and remote fundraising; contacted 30 prospective sponsors and helped raise KSh 800,000 (approximately \$6,000).

Project Leader *Plastiki Rafiki | Nairobi, Kenya***September 2022 – May 2024**

- Led a 22-member student-run NGO that produced and sold recycled-plastic products, generated \$6,500 per quarter, and diverted an estimated 200 kg of plastic waste per month through community collection programs.
- Established four production workshops, each employing 15 local residents.

Team Leader *Dr. Choksey's Albinism Foundation | Nairobi, Kenya***June 2023 – Feb 2024**

- Led a 3-person team to develop a Python-generated QR code donation system and a website to promote the foundation and collect donations.
- Ran a 2-month pilot program at a Carrefour supermarket in Nairobi, generating \$8,000 in donations.

TECHNICAL SKILLS

Design & Modeling | AutoCAD, Revit, SOLIDWORKS, Rhino, Fusion 360, SketchUp, Blender, Adobe Illustrator, Adobe Photoshop**Coding & Analysis** | Python, MATLAB, Microsoft Excel (VBA, Power Query), ArcGIS, HEC-HMS, ETABS, RSMean**Fabrication & Field Work** | Laser Cutting, 3D Printing, Resin Molding, CNC Machining, Plastic Shredding & Extrusion, Power Tools, Surveying Instrumentation**Certifications & Languages** | Fluent in English and Swahili, Red Cross First Aid Certified, International Driver's License